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Roll No. _____

BS/Sem V/150

B. Sc. (Hons) Semester V Examination

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PHYSICS

Paper No. BPT-505 : Electromagnetic Theory

Time : 3 hours

Full Marks : 70

(Write your Roll No. at the top immediately on the receipt of this question paper)

The figures in the right-hand margin indicate marks.

Answer all questions

All symbols have their usual meanings unless specified otherwise

1. Answer any four of the following :

4×5

- (a) If the force between the source charge q and the test charge Q is given as

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \left(1 + \frac{r}{\lambda}\right) e^{-r/\lambda} \hat{r}$$

Calculate the potential due to this point charge q . Is it equivalent as $V(r) = \frac{1}{4\pi\epsilon_0} \frac{q}{r} e^{-r/\lambda}$?

- (b) Obtain a nontrivial vector function that has zero divergence and zero curl everywhere.
- (c) What is the electric field inside and outside of a long hollow cylinder which carries a uniform surface charge σ ?
- (d) A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$. Calculate the cutoff frequency of TM_{11} mode.
- (e) The dielectric constant of pure water is 80. Determine the Brewster angle for parallel polarization and the corresponding angle of transmission.
- (f) In a non-magnetic medium $\vec{E} = 4 \sin(2\pi \times 10^7 t - 0.8x) \hat{a}_z$ V/m. Compute the time average power carried by the wave.

— (2) —

2. (a) Using two-dimensional Laplace equation, obtain the potential $V(x, y)$ for a problem which has two infinite grounded metal plates lie parallel to x - z plane, one at $y = 0$ and other at $y = \pi$ and the left end at $x = 0$ is closed off with an infinite strip insulated from the two plates and maintained at a specified potential $V_0(y) = V_0$. **7**
- (b) In electrostatics, show that the boundary conditions for the electrostatic fields show discontinuity whereas that of potential are continuous. **5**

— (3) —

3. (a) A uniform plane wave propagating in a medium has

$$\vec{E} = 2e^{-\alpha z} \sin(10^8 t - \beta z) \hat{a}_y \text{ V/m}$$

If medium is characterized by $\varepsilon_r = 1$, $\mu_r = 20$ and $\sigma = 3 \text{ mhos/m}$. Find α and β and $\vec{H}$. 6

- (b) A plane e.m. wave incident normally on the boundary $z = 0$ between a perfect dielectric (medium 1) and a perfect conductor (medium 2). Show that there is formation of standing wave in medium 1. 6

— (4) —

4. (a) What are bound charges and bound currents? How do the Maxwell's equation modify due to such charge and currents? 7
- (b) Consider a parallel plate capacitor immersed in sea water carries a voltage $V_0 e^{j\omega t}$. If the sea water is characterized by a permittivity $\varepsilon = 81\varepsilon_0$ and permeability $\mu = \mu_0$, resistivity $\rho = 0.23 \Omega \cdot \text{m}$ at frequency $\omega = 2\pi \times 10^8 \text{ Hz}$, what is the ratio of conduction current to displacement current? 5

Or

- (a) If $V = 0$ and $\vec{A} = A_0 \sin(kx - \omega t) \hat{a}_y$ (A_0 , ω and k are parameters). Calculate $\vec{E}$ and $\vec{B}$. Do $\vec{E}$ and $\vec{B}$ satisfy Maxwell's equation in vacuum? If so, how are ω and k related? 7
- (b) Given vector, $\vec{F} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$, can it be written as a gradient of a scalar or as a curl of a vector functions? If so, obtain the function. 5

— (5) —

5. (a) Show that for a rectangular waveguide

$$u_p = \frac{u'}{\sqrt{1 - \left[\frac{f_c}{f}\right]^2}} \quad \text{and} \quad \lambda = \frac{\lambda'}{\sqrt{1 - \left[\frac{f_c}{f}\right]^2}}$$

where $u' = 1/\sqrt{\mu\varepsilon}$ and $\lambda' = u'/f$. 8

- (b) Calculate the dimensions of an air-filled rectangular waveguide for which the cutoff frequencies for TM_{11} and TE_{03} modes are both equal to 12 GHz. At 8 GHz, determine whether the dominant mode will propagate or evanesce in the waveguide. 6

Or

- (a) A y -polarized uniform plane e.m. wave characterized by $(\vec{E}_s, \vec{H}_s)$ with a frequency 100 MHz propagates in air and impinges normally on a perfectly conducting plane at $x = 0$. Assuming the amplitude of $\vec{E}_s$ to be 6 (mV/meter), write the expression for—

- (i) $\vec{E}_i$ and $\vec{H}_i$ of incident wave;
- (ii) $\vec{E}_r$ and $\vec{H}_r$ of reflected wave;
- (iii) $\vec{E}_t$ and $\vec{H}_t$ of transmitted wave.

6

- (b) Find the relation between reflection and transmission coefficients for an obliquely incident e.m. wave in which wave electric field is parallel to the plane of incidence.

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Parameters to be used in the calculations

Given $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}^2$, $\varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

Charge of electron $e = 1.6 \times 10^{-19} \text{ Coulomb}$

External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>